Teacher Wu Kuan | Exam at a Glance #22 — Combinations: Choose, Don't Order

English lecture notes · YouTube Short JCeAolxykKg

1. The trap question

Choose 2 people from 5: 20 or 10?

2. The method

• Ordered (positions, first/second) → permutation P(n, k).
• Just a group → combination C(n, k) = P(n, k) ÷ k!, since each group was ordered k! times.
• “At least one” → total − none.

3. Parent check method

Ask: “Is Amy-and-Ben a different team from Ben-and-Amy?” If not, divide.

Practice

1. Choose a 3-person group (no roles) from 7 people. How many ways?
2. Choose a class president and vice president from 7 people.
3. 8 points lie on a circle. How many triangles use 3 of them as vertices?
4. From 4 boys and 3 girls choose 3 people with at least 1 girl.
5. Pick 3 different numbers from 1–10. What is the probability their sum is even?

Answers

1. 35
2. 42
3. 56
4. 31 (35 − 4)
5. 1/2 (60/120)

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